**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

**Model Question**

# MG1DSCPEG100- FUNDAMENTALS OF PETROLEUM GEOCHEMISTRY

Duration: 1.5 hrs. Maximum marks: 50

*Students should attempt at least one question from each course outcome to enhance their overall outcome attainability*

## Part A

Multiple Choice Questions
Answer **All** Questions
Each question carries **1** mark

1. Which of the following processes is most associated with diagenesis? [Remember] [Course Outcome 1]
   a) Biodegradation b) Hydrocarbon migration c) Thermal cracking d) Pyrolysis
2. Which of the following is a primary factor influencing the occurrence of petroleum? [Understand] [Course Outcome 1]
   a) Atmospheric pressure b) Geological formations c) Ocean currents d) Solar radiation
3. Which of the following is a type of structural trap? [Remember] [Course Outcome 2]
   a) Fault trap b) Sandstone lens c) Carbonate reef d) Shale formation
4. Which of the following is a direct indicator of hydrocarbons during geological exploration? [Remember] [Course Outcome 3]
   a) Anticlines and synclines b) Salt domes c) Oil seeps d) Fault lines
5. Name a non-invasive method used for oil and gas exploration, especially in environmentally sensitive areas. [Remember] [Course Outcome 3]
   a) Core Sampling b) Seismic Survey c) Remote Sensing d) Magnetic Logging
6. Which logging method is particularly useful for detecting the presence of hydrocarbons in a reservoir? [Remember] [Course Outcome 3]
   a) Density Logging b) Gamma-Ray Logging c) Resistivity Logging d) Electromagnetic Logging
7. Select which of the following hydrocarbon is primarily found in natural gas? [Remember] [Course Outcome 4]
   a) Benzene b) Ethylene c) Methane d) Cyclohexane

8. Coal bed methane (CBM) is primarily derived from: [Understand] [Course Outcome 4]
   a) Coal seams b) Natural gas reservoirs c) Oil fields d) Gas hydrates
9. Low aniline point of hydrocarbon typically indicates: [Remember] [Course Outcome 5]
   a) High aromatic content b) Low volatility c) High paraffin content d) High density
10. Identify which of the following instruments is used to measure refractive index? [Remember] [Course Outcome 5]
    a) Bomb calorimeter b) Refractometer c) Flash point tester d) Thermometer

**(1x10=10)**

## Part B

Short Answer Type Questions
Answer **4** Questions
Each question carries **3** marks

11. List out the benefits of deasphalting process. [Analyse] [Course Outcome 1]
12. Illustrate migration of hydrocarbons from carrier beds to reservoir rock. [Apply] [Course Outcome 2]
13. Compile electromagnetic methods with seismic surveys in oil exploration techniques. [Apply] [Course Outcome 3]
14. Explain radioactive logging and discuss how is it used in petroleum exploration. [Understand] [Course Outcome 3]
15. Explain the significance of understanding the composition of oxygen compounds in crude oil. [Understand] [Course Outcome 4]
16. Explain about surface tension, and why is it important for hydrocarbons? [Understand] [Course Outcome 5]

**(3x4=12)**

## Part C

Short Essay Type Questions
Answer **4** Questions
Each question carries **7** marks

17. Justify that modern theory provides better explanation for origin of petroleum than other theories. [Evaluate] [Course Outcome 1]
18. Discuss the classification of structural trap [Remember] [Course Outcome 2]

19. Illustrate how does core sampling enhance the understanding of subsurface geology. Discuss its application in oil exploration. [Analyse] [Course Outcome 3]
20. Describe the importance of geochemical exploration in oil prospecting [Understand] [Course Outcome 3]
21. Discuss about Unconventional resources of hydrocarbons [Understand] [Course Outcome 4]
22. Discuss in detail about viscosity and viscosity reducers. [Understand] [Course Outcome 5]

**(7x4=28)**